A.Irama: From Brainwaves to Composition

Sep 28, 2026 · @JeffreyZ

Three people's one-minute resting brainwaves became a four-minute piece for Gendang, Rebab, Serunai and Gong, shaped like the opening of a Main Puteri performance. The EEG decides the melody, the drumming density and when the second voice sings; tradition decides the form and the gong cycle.

What we did

The work ran in two stages: first we measured and mapped the data honestly, then we composed from it.

1. **Loaded the three recordings.** Subjects 01–03, eyes-open resting baseline (run R01) from the PhysioNet EEG Motor Movement/Imagery dataset: 64 channels at 160 Hz, 60 seconds each. We kept the six A.Irama sites: Fp1, Fp2, AF7, AF8, T7, T8.
2. **Analysed them the way the thesis describes.** 2-second windows moving in 250 ms steps; band power per instrument band (delta, theta, alpha, gamma) and frontal alpha coherence (C_fAlpha) between AF7 and AF8.
3. **Voiced the four instruments in Ableton Live.** No Kelantan recordings were available, so each instrument is a built-in stand-in: tabla samples for the Gendang, a bowed-string model for the Rebab, a reed-like synth for the Serunai, a resonator model for the Gong.
4. **Played the exported MIDI and built a Brainwave Map for each subject.** An interactive page lines up raw EEG, spectrum, the four instrument bands and coherence with the MIDI and the recorded audio, bar by bar.
5. **Audited the export.** The map showed the Python export rotates notes round-robin across the tracks, uses two or three pitches, and follows time rather than the brain. That finding went into the thesis appendices.
6. **Researched the tradition.** Mak Yong and Main Puteri sources gave the opening order (bertabuh, then the salutation to the rebab), the 16-beat gong cycle, the ibu–anak drum conversation and rebab ornamentation.
7. **Composed a new piece from the EEG.** A composition engine reads each subject's brain signals and writes the Rebab melody, Gendang density and Serunai entries, inside a traditional form. Subjects 01 → 02 → 03 become three movements between an opening and a closing.

What the data showed

The three people's brains were strikingly different at rest, and that difference is what gives the piece its three characters.

Subject	Frontal alpha coherence (median)	What stands out	Character in the piece
Subject 01	0.62	Moderate blinks; the cleanest recording	Calm, steady
Subject 02	0.43	Temporal gamma about 3× the others, most likely jaw or temple muscle tension	Restless, searching
Subject 03	0.90	Eye activity in 98% of windows; coherence moves with it, so it is likely inflated	Intense, the climax

These are single one-minute recordings of healthy volunteers, not A.Irama listeners. They describe the three people, not an effect of the system.

From data to music

Each instrument listens to the part of the brain the thesis assigns it in the Cultural Timbre Mapping; the tradition supplies how that instrument would really behave.

Instrument	Listens to	Electrodes	What the data changes	How it is made musical
Rebab	Alpha, 8–13 Hz	AF7, AF8 (frontal)	The melody: its rise and fall follows the subject's frontal alpha, sampled every second and smoothed over 4 s	Kept within a fifth on D (D E F G A), moves step by step like a voice, notes held and joined by slides, grace notes, turns and a breath before each gong
Gendang	Theta, 4–8 Hz	All six	How busy the drumming is in each 8-second gong cycle: sparse, medium or dense	Two drumheads in conversation, ibu (low) and anak (high), in a–b–a1–b1 phrases that land with the gong

Instrument	Listens to	Electrodes	What the data changes	How it is made musical
Serunai	Gamma, 30–45 Hz	T7, T8 (temporal)	Whether it sings in each gong cycle	Shadows the Rebab an octave higher and slightly behind, the way a voice follows the rebab in Mak Yong
Gong	Not driven by data	Fp1, Fp2 in the thesis	Nothing; it is the fixed frame	The Mak Yong 16-beat cycle: high gong on beats 4, 8, 12 and 14, low gong on 16

Thresholds are shared across all three subjects, so a subject with more theta really does get denser drumming. The contrasts between movements are measured, not written in.

The composition

The piece runs 114 bars at 120 BPM, about 3 minutes 48 seconds, and follows the arc of a ceremony: an opening, three people, and a return to rest.

Section	Bars	Source	What happens	The story
Bertabuh	1–8	Composed	Gong cycle begins, Gendang enters, Rebab tunes on A then settles on D	The stage is opened, as the ensemble does before a Main Puteri
Movement 1	9–40	Subject 01	Medium drumming, Rebab alone, no Serunai	A calm mind: one voice, steady ground
Movement 2	41–72	Subject 02	Sparse drumming; the Serunai shadows the Rebab in every cycle	A restless mind: the second voice will not settle
Movement 3	73–104	Subject 03	Densest drumming with fast anak strokes; Serunai in half the cycles	The climax, the gerak angin, the winds stirring
Penutup	105–114	Composed	Rebab descends A–G–F–E–D, drums fade, final low gong at bar 113	Resolution, the semangat restored

Every movement is 32 bars, exactly eight gong cycles, and each cycle carries eight seconds of that subject's EEG. The Rebab ends every movement on D, so the three people hand the melody to each other.

What is data and what is composed

Say "composed from the brainwaves", not "the brainwaves playing": the data shapes the music, and a composer chose how.

From the EEG: the Rebab's contour, how busy each gong cycle of drumming is, when the Serunai sings, and the order and contrast of the three movements.

Composed: the key (D, within a fifth), the ornaments, the 16-beat gong cycle, the drum patterns, the opening and closing sections, and the 120 BPM tempo.

Four points to be upfront about:

- **This is not the live A.Irama system.** The recordings are pre-recorded volunteers who never heard the music; nothing here shows entrainment.
- **The instruments are stand-ins.** Ableton models, not the FFT-validated Kelantan recordings the thesis specifies.
- **Subject 03's high coherence is probably eye activity.** It is shown as it came out, and named as such.
- **The raw Python export was not used in the piece.** It rotated notes across tracks regardless of the brain; the Brainwave Maps show why a new mapping was needed.

Narrating it

Shape the talk like the piece itself: open the stage, meet three people one by one, then resolve. About 15 minutes plus the full performance at the end.

1. **Bertabuh: open the stage (1 min).** Show: the title and a photo of a Main Puteri ensemble. Say: "In Kelantan, a healing ceremony begins by opening the stage with music. Tonight we open it with three people's brainwaves." Play bars 1–8.
2. **The question (1 min).** Show: one line, "Can the brain lead a traditional Malay ensemble?" Say: A.Irama maps brain activity to Gendang, Rebab, Serunai and Gong, and aims to guide listeners toward calm, synchronised states.
3. **Where the data comes from (1 min).** Show: the six electrode positions on a head diagram. Say: three volunteers, one minute each, eyes open and at rest, from a public research dataset.

4. **What a brainwave looks like (2 min).** Show: the Subject 01 Brainwave Map, live. Scroll the raw EEG, point out a blink, then the four instrument bands. Say: "Each instrument listens to one part of the brain."
5. **The mapping (2 min).** Show: the four-row table from "From data to music". Say one line per instrument: Rebab = frontal alpha, the mind's calm; Gendang = theta; Serunai = temporal gamma; Gong = the ritual frame.
6. **The honest detour (1 min).** Show: the Map's MIDI rows rotating across tracks. Say: "Our first export ignored the brain; the visualisation caught it. So we rebuilt the mapping." Audiences trust a process that shows its corrections.
7. **Learning from the tradition (2 min).** Show: the 16-beat gong cycle drawn as a circle, high gong on 4, 8, 12, 14 and low gong on 16. Play a short Mak Yong or Main Puteri reference clip. Say: the gong frame, the drum conversation and the rebab's ornaments come from here.
8. **Three minds, three movements (2 min).** Show: the subjects table from "What the data showed". Play 15–20 seconds of each movement back to back. Say: "Same instruments, same rules; the difference you hear is the difference between three brains."
9. **What is data and what is art (1 min).** Show: the two lists from "What is data and what is composed". Say: the brain shapes the music; a composer chose how.
10. **What comes next (1 min).** Show: real Kelantan instrument recordings, the closed-loop live study, practitioner review. Say: in the live system, the music changes as the listener's brain responds.
11. **The performance (4 min).** Show: the Brainwave Maps on screen, switching at each movement. Play the full piece from bar 1. Say nothing; let the Penutup and the final gong close the session.

For slides 4, 5 and 11, keep the maps open in browser tabs. Export the piece as a 24-bit WAV first so playback doesn't depend on Ableton.

Sources

- Brainwave Maps: Subject 01, Subject 02, Subject 03. Each plays its own raw-data recording, so mute the page during the performance.
- EEG data: PhysioNet EEG Motor Movement/Imagery Dataset, subjects S001–S003, run R01.
- Mak Yong Music of Malaysia: Negotiating Complex Musical Content, Malaysian Journal of Performing and Visual Arts: gong cycles, drum patterns, rebab ornamentation.
- Main Puteri — Poetics and Politics, Izzuddin Ramli: the ensemble and the bertabuh opening.

- Indigenous Traditions: Main Puteri and Mak Yong, TEAK Asian Traditional Theatre & Dance.
- Listening: Main Puteri (YouTube), Rebab Kelantan (YouTube), Menghadap Rebab (SoundCloud).